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Deciphering ADFGVX messages from the Eastern Front of World War I

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Abstract

In the last months of the World War I (WWI), the German Army and diplomatic services used the ADFGVX hand-cipher system to encrypt radio messages between Germany and its outposts and stations in the Balkans, the Black Sea and in the Middle East. Hundreds of cryptograms were intercepted from July to December 1918 by British and US military intelligence, who were able to recover most of the keys, and decipher most of the cryptograms using manual cryptanalysis methods. Fortunately, the original cryptograms have been preserved by James Rives Childs, the US officer assigned to G.2 A.6, the SIGINT section of American Expeditionary Forces (AEF) in Paris, and they appear in his book, General Solution of the ADFGVX Cipher System, published by Aegean Press Park in 2000.

In this article, the authors present the results of an effort towards the complete cryptanalysis of the messages, and an analysis of their contents. The authors present a new computerized method for the ciphertext-only cryptanalysis of ADFGVX messages which they developed for that purpose. They also provide details on how all the keys were recovered and almost all of the messages decrypted, despite the low quality of significant parts of the intercepted material.

The analysis of the messages in their historical context provides a unique insight into key events, such as the withdrawal of the German troops from Romania, and the impact of the Kiel Mutiny on communications. Both events had major political and military consequences for Germany in the Eastern Front. Cryptanalysis allowed the Entente Powers to closely monitor those events as well as others developments, also highlighting inherent weaknesses in German cryptographic and cryptanalytic capabilities. The authors plan to publish online all the decrypted messages.

Keywords: World War I, Eastern Front, ADFGVX, Ciphertext-Only Attack, Transposition, Substitution, Fractionation, Georges Painvin, James Rives Childs, Hill Climbing

1 Introduction

The ADFGVX cipher was designed by Fritz Nebel and introduced by the German Army in June 1918, as an extension to the ADFGX cipher, introduced three months earlier. The ADFGVX cipher was the most advanced field cipher of its time, as it combined fractionation, substitution, and transposition. The story of its successful codebreaking by the French cryptanalyst Georges Painvin (1886–1980) is well known [DL98, Kah96]. The decipherment of one of the messages, known as "Le Radiogramme de la Victoire", provided the Entente allies with critical details about an upcoming major German offensive in the beginning of June 1918.

Less well known, however, is the use of the ADFGVX cipher in the Eastern Front, for strategic communications between Berlin and several outposts in the East such as Romania, the Turkish Empire, and the German Military Mission in the Caucasus. This ADFGVX traffic was intercepted by British and American listening posts from July to December 1918. It was partially decrypted by James Rives Childs (1893-1987), an officer at the American Expeditionary Force G.2 A.6 section, who later wrote several reports on his work [Chi00, Chi69, Chi19, Chi35]. He also preserved logs of ADFGVX traffic intercepted between September and December 1918. The cryptograms were published by Aegean Press Park in 2000, after Childs's death, as an appendix to Childs's book about the ADFGVX cipher [Chi00]. The book contains 460 intercepted German radio messages, some sent as multipart cryptograms, with a total of 668 cryptograms. We were able to decrypt 618 (93%) of those 668 cryptograms with our new methods, and transcribe them into a readable German text.

From the historical point of view, this collection of German ADFGVX cryptograms provides an unique insight into Allied Signals Intelligence (SIGINT) and cryptanalytic capabilities towards the end of the war [Chi00]. According to David Kahn, the five US interception posts alone "snatched 72,688 German messages from the airwaves" from the fall of 1917 until the end of the conflict [Kah96, p. 334]. Unfortunately, most of the intercepted material has not been preserved. In his study on the British Army and Signals Intelligence during WWI, the Canadian historian John Ferris mentions that the British Army destroyed "all but a handful of its original intelligence files" [Fer88, p. 24]. For instance, only 25 files recording the activities of British codebreakers operating in France have survived, out of 3330. Similarly, the German Imperial Navy destroyed nearly all of its original documents for that period. German Army records are also very scarce, due to extensive destruction during and after World War II.

The analysis of the Eastern Front ADFGVX messages poses major challenges for modern researchers, in addition to the technical cryptanalytic challenges. In his study Ferris quotes the British intelligence branch in Iraq complaining that "the mass of information received by any G.H.Q. almost came in disconnected fragments of very varying value ... fragments were the general rule, and the bulk of intelligence work at any G.H.Q really consists in putting together a gigantic jigsaw puzzle from innumerable little bits, good, bad and indifferent" [Fer88, p. 24]. Wartime intelligence work - cryptanalysis and analysis, is usually done under high time pressure, and its outcome may have serious consequences, which is not the case for modern research. But despite extensive prior research about the WWI (1914-1918) and the availability of recently declassified documents, the authors of this article still needed to face several major and often similar challenges. Most of WWI historical research does not cover the period after November 11, 1918, after the German delegation signed the Armistice with the Entente in Compiègne, while a large part of the ADFGVX traffic is from that period. Furthermore, most of the prior historical research focuses on the Western Front, as developments on this front eventually decided the outcome of the war. Historians often refer to the Eastern Front as the "forgotten front".

Childs's collection of ADFGVX messages is just a small part of a "gigantic jigsaw puzzle". The interception logs preserved by Childs are organized chronically by their transmission or interception time, rather than by topic. From the geographical perspective they cover a vast area from the Baltic Sea to the Ottoman Empire, including Palestine and Northern Iraq, Eastern Europe (especially Romania), the Black Sea area, and Georgia. The messages deal with tactical movements of German troops in the Caucasus, Romania and the Ottoman Empire, with technical, logistical, and communications matters, as well as with the political and strategic situation in Germany and in its outposts. Each topic requires intensive research, including the review of historical archives, to establish the historical context and reconstruct the meaning and implications of the various messages.

After we deciphered the messages, our next task was to reconstruct their structure and order, and to collate multipart cryptograms into complete messages. After that, we tried to reconstruct a readable German text for each message, cleaning up garbles. Next, we made an effort to identify the callsigns, acronyms, persons, and entities involved in the communications. This was an iterative process as the interpretation of callsigns often changed as more messages

were analyzed. The last step was to bring the contents of the messages within their historical context. This required a thorough review of the history of the war on the Eastern Front, of the history of Germany's communications systems and procedures, and of German intelligence organizations. The latter topics are still limited to experts and mainly focused on one unit, Abteilung IIIb [Pöh05, p. 25–54]. For all those reasons the historical part of this article cannot claim completeness.

The remainder of this article is organized as follows. In Section 2, we provide a description of the ADFGVX cipher. In Section 3, we provide a review of historical cryptanalytic methods, as well as modern methods. In Section 4, we present a new computerized method we developed for this research, and in Section 5, we describe the process of how most of the messages were decrypted. In Section 6, we present an analysis of German and Allies' cryptographic and cryptanalytic capabilities, as well as a profile of James Rives Childs, and the analysis of several key historical events reflected in the deciphered radiograms. Finally, in Section 7, we conclude our findings from the technical and historical perspectives.

2 Description of the Cipher

The cipher is named after the six Morse symbols, A, D, F, G, V, and X, the only symbols used when transmitting ADFGVX messages. Those symbols were deliberately chosen in order to reduce reception errors, as they sound different when transmitted in Morse (see Figure 1 for the Morse symbols). An earlier version of the cipher, the ADFGX, only used five symbols. We describe here the working principle of the cipher and an analysis of the size of its keyspace.

A:	. -	D:	- . .	F:	
G:	- - .	V:	. . . -	X:	- . . -

Figure 1: ADFGVX Morse symbols

2.1 Working Principle of the ADFGVX Cipher

Encryption consists of two phases, the substitution phase, and the transposition phase. We illustrate the process using an original ADFGVX message from November 1918, with the following plaintext:

FUERxLEGATIONxALLEMANDExKONSTANTINOPELx

In the substitution phase, we replace each letter of the plaintext with a pair of ADFGVX symbols, according to a Polybius square. In the early version of the cipher, ADFGX, the size of the square was $5 \cdot 5 = 25$, in which all the letters of the alphabet appeared, except for the letter J which was replaced by the letter I. The main drawback of this early version was the need to spell out numbers in words, resulting in longer messages and more frequent repetitions. This issue was addressed in the ADFGVX cipher, by adding V as the 6th symbol. The Polybius $6 \cdot 6 = 36$ square for ADFGVX allows the encryption of all 26 letters of the alphabet, as well as the ten digits, 0 to 9.

In our example, we use the key in effect in the Eastern Front from November 7 to November 9, 1918. The Polybius square for this key is shown in Figure 2.

We start with the first plaintext letter, F. We first locate F in the square and it is at the intersection of row X and column Y. We, therefore, replace it with the pair XX. Similarly, the second plaintext letter U is replaced by AV, the third letter E by DV, and so on. Note that in each pair, the first ADFGVX symbol represents the row, and the second one represents the column. We obtain the following interim text:

XX AV DV AD VV DD DV DG XD GV GG FX XG VV XD DD DD DV AF XD XG XV DV VV VG FX XG
DX GV XD XG GV GG XG FX AA DV DD VV

	A	D	F	G	V	X
A	P	R	M	Y	U	N
D	3	L	Z	G	E	S
F	8	C	7	1	Q	0
G	V	2	9	I	T	B
V	4	0	6	K	X	H
X	5	A	J	N	D	F

Figure 2: Polybius square - Eastern Front ADFGVX key for November 7-9, 1918

6	12	7	15	1	11	16	5	8	14	3	18	9	13	2	17	20	10	19	4
X	X	A	V	D	V	A	D	V	V	D	D	D	V	D	G	X	D	G	V
G	G	F	X	X	G	V	V	X	D	D	D	D	D	V	A	F	X	D	
X	G	X	V	D	V	V	V	V	G	F	X	X	G	D	X	G	V	X	D
X	G	G	V	G	G	X	G	F	X	A	A	D	V	D	F	V	V		

Figure 3: Transposition - Interim rectangle

Next we apply a columnar transposition to the resulting interim text. In our example, the transposition key in effect during November 7-9, 1918, is as follows:

6, 12, 7, 15, 1, 11, 16, 5, 8, 14, 3, 18, 9, 13, 2, 17, 20, 10, 19, 4

This key has 20 elements. We first inscribe the interim text, obtained after substitution, into an *interim (transposition) rectangle*, row by row. On top of the interim rectangle, we inscribe the transposition key. We obtain the interim rectangle shown in Figure 3. Note that the last row is incomplete, and has only 18 elements.

We now reposition or transpose the columns, according to the transposition key. The first column in the rectangle is repositioned at column 6, the second column to column 12, and so on. After transposing all the columns, we obtain the *ciphertext rectangle* shown in Figure 4.

Next, we extract the ciphertext from the ciphertext rectangle, this time column-by-column, starting with DX DG, then DD DD, then DD FA, then VD DD, then DV VG, and so on. Note that column 4 is incomplete, so we extracted only 3 symbols. Finally, we obtain the following ciphertext:

DX DG DD DD DD FA VD DD VV GX GX XA FX GV XV FD DX DD FV VV GV GX GG GV DG VV DG
XV XV VA VV XG VX DD DX AG XX XA GV

After transposition, the two ADFGVX symbols which represent a plaintext letter most likely will not appear next to each other in the ciphertext. This effect is called *fractionation*. Fractionation significantly increases the security of the cipher.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
D	D	D	V	D	X	A	V	D	D	V	X	V	V	V	A	G	D	G	X
X	D	D	D	V	G	F	X	D	F	G	G	D	D	X	V	V	D	X	A
D	D	F	D	V	X	X	V	X	V	V	G	G	G	V	V	X	X	X	G
G	D	A		G	X	G	F	D	V	G	G	V	X	V	X	F	A		V

Figure 4: Transposition - Ciphertext rectangle

To decrypt a ciphertext, we apply those steps in reverse. We prepare a blank interim rectangle, only marking the transposition key on top of it. Since the length of the ciphertext is known, we can identify which of the cells in the last row, on the right side of that row, will remain empty. We now create a blank ciphertext rectangle and mark the corresponding empty cells. Next, we copy the ciphertext into the ciphertext rectangle, column-by-column. We then apply the inverse transposition on the columns of the ciphertext rectangle, to obtain the interim rectangle. For example, column 1 in the ciphertext rectangle is repositioned to be column 5 in the interim rectangle. Next, we extract the interim text from the interim rectangle, row-by-row. Finally, we divide that interim text into pairs of ADFGVX symbols, and replace each pair with the corresponding plaintext letter, according to the substitution Polybius square.

2.2 Analysis of the Keyspace Size

An ADFGVX key consists of two elements, the substitution Polybius square, and the transposition key. The number of possible unique substitution keys is $36!$, and the number of unique transposition key varies according to the length of the key. Based on an analysis of the keys recovered, the shortest transposition key had 16 elements, and the longest had 23. Therefore, the size of the ADFGVX keyspace is as follows:

$$36! \cdot \sum_{n=16}^{23} n! \approx 10^{64} \approx 2^{213} \quad (1)$$

For ADFGX, the size of the keyspace is as follows:

$$25! \cdot \sum_{n=16}^{23} n! \approx 10^{48} \approx 2^{158} \quad (2)$$

3 Prior Cryptanalysis

In this section, we provide an overview of the historical cryptanalytic methods, as well as modern approaches. The first break into the early version ADFGVX cipher, the ADFGX cipher, was achieved in the spring of 1918 by Georges Painvin, his method addressing only the special cases of messages with similar beginnings or similar endings [DL98, Kah96]. Shortly after the war, towards the end of 1918, James Rives Childs from the US codebreaking unit G.2 A.6 was able to develop a more general solution [Chi00, Chi69, Chi19, Chi35]. Marcel Givierge, the head of the French Bureau du Chiffre, independently developed a very similar method in the early 1920s. In recent years, additional methods have been proposed for the cryptanalysis of ADFGVX/ADFGX. We present here an overview of the historical methods, as well as of the modern approaches.

3.1 Painvin's Methods - Spring 1918

Georges Painvin worked under great pressure, while the German Army launched their 1918 Spring Offensive, to break into the new ADFGX cipher, and into the ADFGVX cipher introduced shortly thereafter. His first achievement was the identification of the details of the system and of its mechanism. On April 6, 1918, Painvin's was able for the first time to recover a daily key and decipher messages encrypted with this key. He achieved that by taking advantage of stereotyped messages with similar endings or similar beginnings, encrypted using the same key.

For illustration purposes, we shall assume two messages have the same ending, and have been encrypted with the same key. Since the two messages share the same ending, their interim text, after the substitution phase (see Section 2) will also share the same ending segment of the interim text. We now look at the interim transposition rectangles, for each one of the two messages. The similar ending segment occupies the last rows of each rectangle. The two rectangles have the same number of columns (also equal to the key length). Therefore, the last fragment of each one of the columns of the first rectangle will be identical to the last fragment of one of the

columns in the second rectangle. Transposition preserves the continuity of columns, and those similar fragment at the end of the columns are also preserved in the ciphertext rectangles, after transposition. For each column in the first ciphertext rectangle, there will therefore be one column in the second ciphertext rectangle, which ends with the same fragment. When extracting the ciphertexts, column-by-column, from the ciphertext rectangles, those identical fragments are preserved and will appear in the final ciphertexts.

An cryptanalyst looking at the two ciphertexts, and suspecting that the original plaintexts have the same ending, will be able to identify those identical fragments of text which appear in both ciphertexts. By analyzing the positions of the matching fragments in the two ciphertext rectangles, the analyst is able to deduce the transposition key. As soon as the transposition key is recovered, and the transposition undone, he just needs to solve a simple substitution cipher, using both messages. The method is described in detail in [Fri96]. A very similar method, for the case of similar beginnings, is also described.

With this method, Painvin was able to recover nine daily keys used in the Western Front in 1918 and a tenth key was recovered by British codebreakers [Chi00, Chi69, Chi19, Chi35]. While they cover only a fraction of the daily keys used in the Western Front in the spring and summer of 1918, those ten keys were used to encrypt 50% of all the traffic during that period [Chi00, Chi69, Chi19, Chi35]. This can be explained by the fact that the availability of large numbers of messages on a given day provided Painvin and his team for more opportunities to find pairs of messages with similar endings or beginnings. In addition, increased traffic by itself was often a sign of a German offensive planned for a few days later.

3.2 Childs's Method - 1918

When working on the decryption of Eastern Front traffic, Childs initially applied Painvin's methods for the special cases of messages with similar endings or similar beginnings. Only after the end of the war was Childs able to develop a more general solution for the recovery of ADFGVX keys. Childs' method is described in a War Department report, "General Solution for the ADFGVX Cipher System" [Fri34], and in Part IV of "Military Cryptanalysis" [Fri96]. The report is reproduced in Childs's "General Solution of the ADFGVX Cipher System" [Chi00], published by Aegean Park Press in 2000.

At the heart of the method is the fact that the frequencies of the symbols A, D, F, G, V and X, when used as the left component of a pair (see Section 2) at the substitution phase, are very different from their frequencies when appearing as right components of a pair. Friedman provides an example of a substitution square, with the relative frequencies of each letter (relative to a total of 1 000), as shown in Figure 5. In the rightmost column (see (1) in Figure 5), the frequencies of the 6 symbols when used as the left component of a pair are shown. In the bottom row (see (2) in Figure 5), the frequencies of the 6 symbols when used as the right component of a pair are shown. With this substitution square, A is much more likely to appear as the left component of a pair than as the right component. Similarly, X is much more likely to appear as the right component of a pair than as the left component.

Childs's method is applied to a series of messages encrypted with the same key. In the example given by Childs and Friedman, the method is illustrated on a series of twelve cryptograms of various lengths, with 108 to 254 ADFGVX symbols each, 2 312 in total. The transposition key has 17 elements.

The process is iterative. At each step, the analyst analyzes some of the cryptograms, tries to separate symbols from the ciphertexts into two distinct classes or groups, either as left/first components of pairs, or as right/second components of pairs. For each group, he computes the relative frequencies of the A, D, F, G, V, and X symbols, and tries to classify sequences of additional symbols, appearing in the same cryptograms or in new cryptograms.

At first, the analyst determines whether the length of the encryption key is odd or even, by classifying the initial symbols of all the cryptograms. After that, he uses cryptograms having the same length, to classify additional symbols in those cryptograms as left or right components. Next, he uses cryptograms having the same number of elements in the last row of their

	A	D	F	G	V	X	(1)
A	T 92	H 34		E 130		F 29	284
D		L 36	O 75	W 16	R 76	S 61	264
F	A 74		B 10		M 25	I 74	183
G		N 79	P 27	G 16		C 31	153
V		D 42		J 2		K 3	47
X	Q 3	U 26	V 15	X 5	Y 19	Z 1	69
(2)	169	217	127	169	120	198	1000

Figure 5: Frequencies of ADFGVX symbols – as right(1) or left(2) symbols of a pair

transposition rectangles, to classify more symbols in those cryptograms. After having classified enough symbols in several cryptograms, the analyst is able to draw conclusions about the original positions, in the interim transposition rectangles, of columns - after transposition - in the ciphertext transposition rectangles. Using an iterative process, he is finally able to recover the full transposition key.

While termed as a "general solution", this method relies on several factors, that may not be present in all cases. It requires a certain amount of cryptograms (probably 12 or more) as well as several cryptograms having the same length. It also relies on the frequency statistics of the ADFGVX symbols being different for the first (left) elements and the right(second) elements of the pairs, as in Figure 5. In [Fri96], an example of a substitution key engineered to hide those differences is presented. In such a case, the method described here would fail.

Interestingly enough, a very similar method is described by Marcel Givierge, the head of the French Bureau du Chiffre [Giv32]. Childs claims to have developed his method independently of Givierge's and that he did not have had access at the time to Givierge's work [Chi00].

In [Fri96], a solution for the much simpler special case of a complete transposition rectangle is also described. A complete transposition rectangle is obtained when the length of the ciphertext is an exact multiple of the key length.

3.3 Konheim - 1985

Konheim's method is presented in an extended abstract [Kon85] as well as in an appendix to Childs's book [Chi00]. It also relies on uneven frequencies of letters in any language, also reflected in the substitution Polybius Square (see Section 2). First, using a statistical analysis of potential digrams generated by juxtaposing columns of ciphertext symbols, Konheim detects whether any two columns were adjacent in the original (interim) transposition rectangle. Such digrams or pairs of ADFGVX symbols are likely to display the frequency characteristics of a monoalphabetic substitution. Next, the substitution must be solved using "standard techniques", which are not detailed in the paper. Last, after solving the substitution, the transposition key may be recovered.

This algorithm is demonstrated on a single and rather long message, with 1 672 plaintext characters. The ciphertext length has therefore 3 344 ADFGVX symbols. The transposition key is rather short and has only 6 elements. This means that the rows of the transposition rectangle have $3\,344/6 = 559$ elements. For comparison, transposition keys used in 1918 had between 15 to 23 elements, and messages had on average 215 ADFGVX symbols, and each row in the

transposition rectangle usually had no more than 5 to 15 elements.

Furthermore, the statistical methods used for the first step of Konheim's algorithm are rather complex, with many special subcases. Those methods rely on having exceptionally long rows in order to produce statistically significant results. Another caveat of Konheim's method is that he does not specify which "standard techniques" may be applied to solve the substitution before the transposition key has been recovered. The techniques commonly used for solving a monoalphabetic substitution cipher are applicable only if the symbols are in place and not transposed.

3.4 Bauer - 2013

Bauer [Bau13] demonstrates a method based on the analysis of the probable locations of high frequency short words such as "THE". This method is demonstrated on an ADFGX message with 680 symbols, and a transposition key of length 20. As 680 is a multiple of 20, this is a special case of a complete transposition rectangle.

First, columns are matched into pairs of adjacent columns, so that one column contains the left symbols of the original interim text pairs of ADFGX symbols, and the second column contains the corresponding right symbols. Next, the order of those pairs of columns is reconstructed, by guessing the pairs of symbols most likely to represent the high frequency letters "E" and "T", and trying to identify possible occurrences of the word "THE". This allows for the recovery of the transposition key. Finally, the substitution is solved.

4 New Ciphertext-Only Attack

The algorithm we developed is based on hillclimbing, and processes a batch of cryptograms believed to have been encrypted using the same key, rather than trying to decipher single messages. Using a divide-and-conquer approach, the algorithm first tries to recover the transposition key. When the transposition key has been recovered, we are left with a simple substitution cipher, which can be solved using a variety of methods [Gai43, Fri96, Yi14, Jak95, FSN93, CD97, Kin94, RAT93, Luc88]. We also used a hillclimbing algorithm to solve the substitution key. Here, we describe in more details the algorithm to recover the transposition key.

As mentioned in Section 2, the transposition part of ADFGVX is implemented using the classical columnar transposition cipher method. Several methods have been proposed and successfully used for the cryptanalysis of columnar transposition ciphers [Mat93, GSN94, Cla98, RCS03, DG03, CR12, Yi14, MG08, LKW16]. All those methods are based on an iterative attempt to reconstruct the original bigrams (pairs of adjacent letters) or trigrams (triplets of consecutive letters) which appeared originally in the plaintext. Those algorithms usually compare the distribution of bigrams and trigrams in a text decrypted using a putative key, to their natural distribution in the language. For example, in English, the trigrams ION and ING are much more likely to appear in a text than the trigrams YFG or KJN. Unfortunately, because ADFGVX also includes a substitution phase (each plaintext letter being converted into a pair of ADFGVX symbols) those methods do not apply. Furthermore, after transposition, the two symbols which composed an original plaintext letter are most likely to appear in different places in the ciphertext, after transposition, and recovering the transposition key is even more challenging. This effect is also known as fractionation, and was deliberately introduced into the ADFGX and ADFGVX cipher methods. The inventors of the ADFGVX cipher were aware of methods to solve columnar transposition ciphers and simple substitution ciphers, but believed that the combination of a transposition cipher and a substitution cipher, with fractionation, would create a much more secure type of hand cipher.

To recover the transposition cipher, rather than matching the distribution statistics of bigrams or trigrams to their distribution in the natural language (in our case German), our algorithm looks at their Index of Coincidence (IC) [Fri79]. More specifically, we look at the IC of bigrams of ADFGVX symbols, which, if correctly juxtaposed, would represent a single plaintext letter. In addition, for the case of transposition keys of even lengths, we also look at the IC of quadruplets (tetragrams) of consecutive ADFGVX symbols, which, if correctly juxtaposed, would

represent a bigram or pair of consecutive letters in the plaintext. Since we do not know a priori the length of the transposition key, we apply this algorithm to all possible lengths, from 15 to 25 elements.

Our algorithm works as follows:

1. Generate a random putative transposition key
2. Decrypt the ciphertext by applying the inverse of the putative transposition key.
3. For each valid transformation (as defined later in this section) on the transposition key, perform the following:
 - (a) Apply the transformation to modify the current putative transposition key and generate a new putative key.
 - (b) Decrypt the ciphertext with this new putative transposition key.
 - (c) Compute the IC of all resulting bigrams of ADFGVX symbols.
 - (d) If the IC of bigrams with this new putative key is higher than the IC obtained with the key before the transformation, keep that new key. Otherwise, discard the new key, and repeat 3.a, checking the next transformation.
4. As long as the IC can be improved using any of the transformations on the transposition key, repeat Step 3.
5. Otherwise, if the best IC so far is above a certain threshold, the algorithm has probably obtained the correct transposition key, and it tries to solve the substitution key.
6. If the best IC was below the minimal threshold, or if the solution of the substitution key (in Step 5) did not yield a readable and valid plaintext, restart from Step 1.
7. If the Steps 1 to 6 have been performed 1000 times without success, stop. The chosen length of the transposition is probably incorrect, or the cryptograms may be too short, or with too many errors, or may not all belong to the same key.

A critical component of this algorithm is the type of transformations used at each iteration, and applied on the transposition key. Note that the algorithm always tries all possible and valid transformations, rather than a random subset of transformations (as typically implemented by other methods such as Genetic Algorithms and Simulated Annealing). The transformations which the algorithm checks at each iteration consist of:

1. Simple swap of any two elements in the transposition key.
2. Swap of any two segments of consecutive elements in the key.
3. Rotating (cyclically) a segment of consecutive elements in the key (at any position, and of any length).
4. Reversing the key, so that the last element becomes the first, and so on.
5. Swapping the elements of every pair in the key. In the case of a key with an odd length, we just skip the last element of the key.

This algorithm performs very well for transposition keys with odd length. In the case of transposition keys with an even length, it will often fail. The reason is that it is not enough to measure the IC of bigrams of ADFGVX symbols if the key has an even length. To illustrate the problem, we consider a simple transposition key with 8 elements: (4, 6, 2, 3, 7, 5, 1, 8). At some stage of hillclimbing, we may have obtained the putative key (7, 5, 4, 6, 1, 8, 2, 3). In this putative key, the original **pairs** of key elements, e.g. (4, 6) or (2, 3), have been correctly reconstructed, but the relative order of the pairs is wrong. Despite the wrong order of the pairs of key elements, the IC of ADFGVX bigrams in the text obtained after decrypting with this putative key, will be **equal**

Transposition Key Length	Minimum Number of Cryptograms (n)	Total Number of ADFGVX Symbols	Average Cryptogram Length
16	2	486	243.0
16	2	388	194.0
16	2	402	201.0
17	3	560	186.7
18	5	1000	200.0
19	4	676	169.0
19	2	500	250.0
20	5	770	154.0
20	2	534	267.0
20	4	686	171.5
22	4	926	231.5
22	4	766	191.5
22	3	848	282.7
23	2	566	283.0

Table 1: Minimum number of cryptograms required for complete key recovery

to the IC obtained by decrypting the ciphertext with the correct transposition key. This is due to the fact that the decryption with the wrong putative transposition key is nevertheless able to correctly place next to each other the two symbols of each one of the pairs of original ADFGVX symbols. In other words, all the original ADFGVX bigrams have been recovered, albeit not in the same order. A major result of this phenomena is that measuring the IC of ADFGVX bigrams is not enough for keys with an even length. Therefore, in addition to computing the IC of ADFGVX bigrams after decryption with a putative transposition key, our algorithm also measures the IC of quadruplets of ADFGVX symbols, and adds those two ICs to form a combined scoring method.

To illustrate the performance of the algorithm, we performed the following experiment. After dividing the cryptograms into groups according to the keys they were encrypted with, we ran the algorithm on the first n cryptograms of each group, starting from $n = 10$, and each time reducing n until the algorithm failed to recover the key. Rather than selecting the longest or cleanest cryptograms of each group, we just selected the first n cryptograms of each group. Therefore, the n cryptograms selected from each group (key) may significantly vary in terms of length or number of transcription errors. In Table 1, we show the minimum number of n cryptograms required by the algorithm to allow for full key recovery. It can be seen that for 6 keys out of 14, just 2 cryptograms with of a total of 388 to 566 ADFGVX symbols were sufficient to recover transposition keys of length varying from 16 to 23. The worst case was with a key of length 18 which required 5 messages, with a total of 1 000 ADFGVX symbols. In contrast, Childs’s method was demonstrated on a set of 12 cryptograms with a total of 2 312 ADFGVX symbols, for a transposition key of length 17. Konheim’s method was illustrated on a single cryptogram with 3 344 symbols and a very short key with only 6 elements.

5 Deciphering Eastern Front ADFGVX Messages

In this section, we describe the structure and format of the cryptograms. We also describe the step-by-step process used to recover all the encryption keys and decrypt the messages.

5.1 The Messages

The messages were published by Aegean Park Press in 2000, and appear in Appendix B of Childs’s General Solution of the ADFGVX Cipher System [Chi00]. Appendix C of the book also

contains some of the keys for the messages. According to the editor of the book, the messages were provided by Childs twenty years before the publication of the book. The messages appear in the form of interception logs. At the beginning of each page, the following header is shown:

GENERAL HEADQUARTERS, AMERICAN EXPEDITIONARY FORCES
GENERAL STAFF, SECOND SECTION (G.2 A.6) (alp)

"RICHI" ADFGVX CIPHER

(Employed between Berlin and the Black Sea)

Messages Intercepted by British Stations.

Cryptograms from the same date usually appear consecutively. ADFGVX keys for the Eastern Front were in effect for three days. We found, however, a large number of messages included in a day's log but encrypted with unrelated keys. Accordingly to German communications procedures, the maximum length of each single cryptogram could not exceed 250 ADFGVX symbols. Longer messages were to be split into shorter parts - "Teile" in German, often abbreviated as *TL*. Note that this rule was not always followed, and some of the cryptograms are actually longer, some longer than 400 symbols. Each message, composed of one or more parts (Teile), had an header sent in cleartext, as in the following example:

POT, COS v NKJ (3.10 a.m., October 7th)
FÜR COS DEUTSCHE DELEGATO WIRTCHI ABTLG 2305 (6) 2 TLE

The first line contains the callsigns of the receiving stations - POT and COS in this example - followed by a "v" which abbreviates the German word "von" (from) indicating the sender, in this case NKJ. On this line the operators of the Allied interception station usually added some information, such as the date and time of reception. The second line usually contains (in clear-text) the name or location of the recipient of the message. In this example the recipient is the Economic Department of the German Delegation at Tiflis (COS), the capital of Georgia. "2305" refers to the transmission hour (11:05 p.m.) and "(6)" to the transmission day. "2 TLE" indicates that the message consists of two parts (Teile).

Each part of a multipart message consists of a separate cryptogram, with its own header. An example of such a header is "1 TL RICHI-322" which indicates that this is the first part of the message, it is encrypted using the "RICHI" (codename for ADFGVX as used in the Eastern Front) keys, and this first part consists of 322 ADFGVX symbols. Next, the cryptogram is written in groups of 5 ADFGVX symbols. The second part starts with the header "2 TL RICHI-256", and accordingly, has 256 symbols. From the cryptographic perspective, the most important items are the cryptogram length and the transmission time, which are both sent in clear, as well as the interception time, logged by the operators of the listening station. Also, the fact that the ADFGVX symbols are transcribed in groups of 5 symbols is often useful, in trying to isolate reception or transcription problems.

In total, there are approximately 460 ADFGVX messages, many of which are composed of multiple ADFGVX cryptograms. In total, there are about 668 ADFGVX cryptograms, if we exclude repetitions of identical messages. The earliest message is from September 19, 1918, and the last message from December 7, 1918. The average length of a cryptogram is 215. About 7% are shorter than 100 symbols, 80% between 100 and 300, 13% longer than 300. The shortest cryptogram has 36 symbols and the longest has 500 (13 have more than 400 symbols).

About one third of the cryptograms have either an incomplete transcription or have transcription errors. This does not include garbled letters which can be observed only after decryption. In fact, after decrypting the cryptograms, we found that almost all of them had garbled letters. The most frequent transcription errors, which are visible even before trying to decrypt a cryptogram, include:

- Symbols transcribed as hyphen ("-"). The interception operators were instructed not to guess Morse symbols they were not sure about, so they marked them instead with hyphen

signs. In case of severe interference, whole groups of 5 symbols each were marked as hyphens.

- In certain groups, only 3 or 4 of the symbols are transcribed, instead of the expected 5 symbols. In other groups, 6 symbols are transcribed.
- In some cases, the transcription includes wrong ADFGVX symbols, such as U or B.
- In some cryptograms, major segments, usually the beginning or ending of a cryptogram, are missing, and instead the operator wrote "jammed" or "interference".
- The number of symbols transcribed does not match the length information sent in clear.
- Sometimes the length information sent in clear is an odd number. This is clearly an error, as such a case may not happen in a properly encrypted ADFGVX message.

Later in this section, we describe how we addressed those errors.

5.2 Recovering the Keys

The first step in the process was to scan the cryptograms from the book. Next, we applied OCR, using the open-source OCR package Tesseract [Smi07], and limiting the set of valid characters to A,D,F,G,V,X, and the hyphen sign ("-"). We then compared OCR results with the original cryptograms, corrected OCR errors, and removed portions of text which were not part of the cryptograms, such as clear text.

The next step was to divide the cryptograms into groups of cryptograms likely to have been intercepted on the same day. At this stage we simply ignored all the cryptograms with transcription errors, and used the two thirds of the cryptograms without such errors. For each daily group, we applied our algorithm described in Section 4, testing all key lengths from 15 to 25. With this method, we were able to recover the keys for most of the days. As expected, the same keys were in effect for 3 consecutive days. With those keys, most of the cryptograms from a given day could be decrypted, but some could not. We suspected that some cryptograms intercepted on the same day did not belong to the same key. Other cryptograms had severe errors, either incorrectly encrypted, incorrectly transmitted, or wrongly intercepted or transcribed. However, this did not prevent our algorithm from recovering most of the keys, as it is robust enough to perform even if up to 20% of the cryptograms do not belong to the same key.

Messages for November 1, 2, and 3 constitute a special case. When trying to apply cryptanalysis on all of that messages from those days, the key could not be successfully recovered. We suspected that there may be more than one key involved. We applied the algorithm again on smaller subsets of cryptograms from those days, trying various combinations of cryptograms. We were able to recover two different keys for those days, with some cryptograms decrypting with one key, and others with the second key.

Next, we created a database of all those keys, and tried again to decrypt each one of the cryptograms, trying each one of the keys. We were able to obtain correct decryptions for almost all of the cryptograms which had no transcription errors, about 400 in total. It turned out that a large number of cryptograms were wrongly placed in the logs, and could be decrypted using the key from another day.

5.3 Handling Errors

Next, we processed the cryptograms with transcription errors, or those that could not yet be decrypted with any of the keys, about 250 in total. For those messages which had hyphens instead of symbols, we used the procedure described here. In ADFGVX, every plaintext character is replaced by a pair of ADFGVX symbols, before transposition. If only one of the symbols has been correctly transcribed, and the other marked as hyphen, there are only 6 options for that missing symbol. Therefore, there are only 6 possible options for the original plaintext character, out of the 36 possible options (a to z, 0 to 9). An example of a message transcribed with two

hyphens follows. The two plaintext letters affected are marked with the "!" sign. For both positions, all the 6 possible original plaintext characters are shown. By looking at the missing letter in the context of the surrounding words, the correct characters, S and G, can be recovered.

	!		!
NACHRCWEF	x	CONSTANTZ	ATEITHEUTE
x	400	AU	SSE
B	T	R	I
E	F	E	S
E	T	Z	T
x	M	I	L
M	I	S	S

About 110 cryptograms had minor transcription errors. Those with few hyphen signs could be easily corrected in most cases using the procedure described above. A similar procedure was also applied to cryptograms having groups with less than 5 symbols. For example, if a 5-symbol group was transcribed as DGVX, then we applied the same procedure by checking 5 cases, - DGVX, D-GVX, DG-VX, DGV-X and DGVX-. For a group with more than 5 symbols transcribed, e.g. GDFAXV, we simply tested 6 cases, each time removing one of the 6 letters and selecting the most probable resulting plaintext.

In addition, about 34 cryptograms had a correct length but a high number of hyphens. In most cases, only some of the missing plaintext characters could be recovered using this procedure. Still, large parts of the cryptograms could be read.

For those cryptograms for which the number of symbols transcribed exceeded the length specified in clear, we simply removed an equivalent number of extra symbols, from the beginning of the cryptogram, or from its end. If this did not work, we tested every possible location in the cryptogram. For those cryptograms for which not enough symbols were transcribed, compared to the length sent in clear, the procedure was more complex. We added groups of hyphens at various positions in the cryptogram, and selected the position which resulted in the best plaintext. Then, we tried to recover some of the symbols marked as hyphen, using the method described above. With those tedious and mostly manual procedures, we were able to recover 27 messages with wrong lengths. While usually the result of interception problems, such cryptograms with wrong lengths may also have been the result of erroneous encryption. In such cases, neither the German receiving side, nor the British who intercepted those messages, were able to decrypt them.

About 50 cryptograms could not be recovered, with any of the procedures described above. Out of those 50 cryptograms, 36 had too many groups missing, mostly due to interference, as also indicated by comments such as "jammed" or "interference" in the interception logs. The remaining 14 had a correct length but could not be decrypted using any of the keys. We suspect that most of those 14 cryptograms were probably wrongly encrypted.

5.4 The Final Keys

Next, we ran the cryptanalysis program again, on the 618 cryptograms we were able to recover, after dividing them according to their keys. We used the transpositions keys we had already obtained, and ran only the substitution solver part. The purpose was to obtain a more accurate substitution key, based on a larger number of cryptograms per key.

Finally, we compared the keys we recovered with the keys provided by Childs in the last part of his book [Chi00]. The results are as follows:

- Eight of the fourteen transposition keys we recovered are identical to Childs's keys, with some minor differences in the substitution keys. Those differences are mainly due to the fact our algorithm is unable to determine the exact position of the digits 0 to 9 in the substitution Polybius square. This may be achieved via traffic analysis, rather than cryptanalysis. Traffic analysis, however, requires additional sources of intelligence, for example, to recognize unit numbers. In addition, for some of the substitution keys, Childs left as blanks the symbols he could not accurately identify.

- Two more keys were provided by Childs, but there were errors so that messages could not be decrypted with those keys. In one case, the error was in the transposition key (an element missing), and in the other case, the error was in the substitution square (two rows of the square swapped). Our program was able to recover the correct keys.
- We recovered four additional keys which do not appear in Childs's book. They refer to the period from November 19 to the beginning of December 1918.

The keys are shown Table 2.

6 Historical Analysis

In this section, we present an overview of German cryptographic and cryptanalytic capabilities at the outset of the war, as a context for the conception of the ADFGX/ADFGVX ciphers, the most sophisticated of the field ciphers developed by Germany during WWI. We also provide a short profile of James Rives Childs, the US officer in charge of decrypting Eastern Front ADFGVX traffic. Additionally, we focus on several topics covered by the decrypted messages:

- The political and military situation in Romania in 1918
- The effect of the Kiel mutiny in November 1918 on German communications
- Cryptic messages with a code within a code
- A message related to Karl Wilhelm Hagelin, the father of the cryptographer Boris Hagelin.

6.1 The German Military and Signals Intelligence

In "The Codebreakers", David Kahn writes: "German cryptology goose-stepped toward war with a top-heavy cryptography and no cryptanalysis" [Kah96, p. 263]. In 1914 Germany's Army and Navy were well equipped in terms of wireless equipments, but less aware of the advantages and disadvantages of the very new radio technology. Their autocratic and rigid nature did not encourage critical thinking and initiative at the lower ranks. In its maneuvers before the war the Army had forbidden the radio personnel from intercepting wireless communications of the opposite forces. The Navy even went so far as to punish a seaman who had informed his superiors that he had intercepted and solved an enciphered message sent between two German radio posts. The Kaiser's generals believed that they would win the next conflict with infantry, artillery, and cavalry as they did in the Franco-Prussian War of 1870-71. Communications security and codebreaking were not considered important.

In addition, early-on in the war Germany suffered a major setback on the SIGINT front, with consequences that would prove to be decisive in the years to come. In August 1914, the Royal Navy cut the German telegraph undersea cables. Germany had to reorganize its communications and to rely mostly on wireless communications, exposing those communications to interception by the enemy. Furthermore, the French cryptographers had an advantage over their German counterparts, with a team of highly experienced cryptographers, who had been monitoring German Army traffic during peacetime German Army maneuvers. Their success in deciphering the main German Army code, the ÜBCHI code, played a critical role in the Western Front during the initial phase of the war.

When the war started, the German Army had no signals security discipline so that fast advancing troops often communicated in plaintext. Their strategic level intelligence on the French and Russian armies was poor. The German Great General Staff used its military intelligence organization, the subsection Abteilung IIIb, only for the purpose of gathering tactical information. Germany had limited information on the British Army since only the Navy collected intelligence on the United Kingdom, focusing mainly on the Royal Navy. The Kriegsmarine had its own naval intelligence service, the Marinenachrichtendienst, which used direction finding and limited codebreaking to locate and identify the enemy's warships. The Foreign Office ran a separate

Period	Transposition Key (Length) Substitution Key	Cryptograms Recovered
Sep 19 – 21	12,2,7,20,10,19,1,13,9,18,3,17,21,8,14,4,6,16,11,22,5,15 (22) "D5613Q9KBNO0HY8EISJUTZFCW7VPML2ARG4X"	15
Oct 4 – 6	4,13,3,14,1,16,9,15,5,19,10,18,6,17,7,20,11,21,8,12,22,2 (22) "YN87PJ3WRUCIEO1SKLZX0DFBH6MT9A2QV54G" <i>Note: Substitution key in Childs's book is erroneous</i>	24
Oct 28 – 31	6,15,12,16,5,7,14,4,13,8,11,1,17,2,10,3,18,9 (18) "HI20SXR UWQY8EK7O619CBJAP453FDZTGLMVN"	33
Nov 1 – 3	3,16,4,15,7,12,18,6,17,8,19,1,13,10,2,14,11,9,5 (19) "UILOF9RCZVSX02G7QTD8WNB5JMHEKPY41A36"	100
Nov 4 – 6	7,10,8,14,3,11,16,1,6,13,4,9,15,5,12,17,2 (17) "17WHFLJ5D2UPEXKVZ9O0Q3Y6R8ABGITCMS4N" <i>Note: Transposition key in Childs's book is erroneous</i>	106
Nov 7 – 9	6,12,7,15,1,11,16,5,8,14,3,18,9,13,2,17,20,10,19,4 (20) "PRMYUW3LZGES8C71QOV29ITB40-KXH-AJNDF"	93
Nov 10 – 12	9,12,7,11,3,8,16,6,14,2,10,15,5,13,1,4 (16) "4ARUT1OIFSKN3-BZPVL D-JMXCWHQ2E-G0-Y-"	46
Nov 13 – 15	13,8,6,16,7,18,1,14,9,20,10,15,17,2,3,11,5,19,4,12 (20) "JZLH-R-S-T-MKDWU-V-B-P-FAO-GIX-CNE"	13
Nov 13 – 15	4,11,5,14,9,7,16,1,12,15,6,10,3,13,8,2 (16) "H-BMUF15PX0DJLR—S6VONKZ-AWITEGC-"	52
Nov 16 – 18, Dec 7 – 9	7,12,1,14,8,16,13,9,19,3,15,4,10,18,6,2,11,17,5 (19) "WG-EITNHUB2R-FDZJS—PYVQL-1OAXMKC"	36
Nov 19 – 21	13,20,3,16,7,14,4,12,8,11,5,15,2,18,17,10,19,6,1,9 (20) "LC58QH7VI2YB9EURO60GX3MTFAKP1D4NJZSW" <i>Note: New key, does not appear in Childs's book</i>	12
Nov 22 – 24	6,12,16,7,14,22,11,18,1,15,8,10,20,2,13,21,3,17,19,5,9,4 (22) "QNZ72XS4C0IJY3RBEKL9FD6GMTHUVWA5O8P1" <i>Note: New key, does not appear in Childs's book</i>	26
Nov 25 – 28	21,9,6,14,10,20,1,16,18,7,15,4,11,22,5,17,23,2,12,8,19,3,13 (23) "HQ05DKZAOYM6BEIWTJ7PSCFLV94132NGURX8" <i>Note: New key, does not appear in Childs's book</i>	33
Nov 28 – Dec 1	9,3,14,10,2,8,15,4,16,11,5,13,6,12,1,7 (16) "782GPY5OQHF91UDNI364TLVXEAR0JZBKMC SW" <i>Note: New key, does not appear in Childs's book</i>	29
	Total:	618

Table 2: List of ADFGVX keys used in the Eastern Front

intelligence network in its embassies and consulates, usually working from open sources. It had its own cipher bureau which did not cooperate with its military counterparts, and until 1918 the Foreign Office still considered its own diplomatic code as unbreakable. While the War Ministry in Berlin was responsible for developing codes and ciphers for the German Army, there was no centralized cryptological institution responsible for analysing their cryptographic security.

At the battle of Tannenberg against a more powerful Russian Army in 1914, general Paul von Hindenburg and his general quartermaster Ernst Ludendorff owed much of their victory to two amateur codebreakers, lieutenant Alexander Bauermeister and professor of Philosophy Ludwig Deubner. Deubner and Bauermeister were attached to the High Army East Command and they collaborated with the cipher bureau of the Austro-Hungarian MILINT, the Evidenzbüro. Early organized Signals Intelligence (SIGINT) in the German Army remained confined to the Eastern Front. Only later, when Hindenburg and Ludendorff headed the Supreme Army Command, the Oberste Heeresleitung (OHL), did they start to integrate SIGINT at the strategic planning level. The Navy followed in February 1916 with its own signal intelligence service and opened an interception and codebreaking center in Neumünster. Until that time most codebreaking successes by the Germans were the result of either individual initiatives or of errors by their enemies.

A major reorganization occurred in the spring of 1917. The "Chief of Field Telegraphy", attached to the General Quarter Master, was renamed "Nachrichtenchef", which Childs translated as "Chief Signal Officer" [Chi35, p. 41]. He was responsible for all communications via phone, telegraph, wireless, and also for the use of codes and ciphers.¹

At the strategic level, the General Chief of Staff of the Field Army, responsible for operational planning, retained its control over Abteilung IIIb. This department, in addition to classical military intelligence from SIGINT and other sources, also acted as political police inside Germany and as a press censor. The evaluation of intelligence gathered from SIGINT about the enemy's armies went directly to the Nachrichtenabteilung (intelligence department), renamed Abteilung Fremde Heere (Foreign Armies Department) in 1917, and having two sections, West and East.

According to former cryptologist Hermann Stützel the E-Stelle (or Entzifferungs-Stelle - Deciphering Station) at the OHL (Supreme Army Command) had three sections [Stü69, p. 543]. The Department of Foreign Armies evaluated the intelligence collected by interception posts. They focused on the enemy's trench codes, direction finding, and reconstructing the enemy's order of battle. The second section dealt with developing field codes for the German Army. A third section monitored diplomatic traffic and worked on cracking diplomatic codes and ciphers. The latter was headed by Hermann Stützel. In 1918, Stützel took the initiative of monitoring traffic between the Foreign Ministry and its embassy at Madrid, and exposed the vulnerability of German diplomatic codes. Despite the scandal following publication of the Zimmermann telegram that brought the USA into the war against the Central Powers in 1917, and despite Stützel's work, the Foreign Ministry kept its conviction that its codes were unbreakable. The Foreign Ministry believed that the Zimmermann message could only have been read by the enemy because they were helped by a traitor [Kah96, p. 294].

Such confidence about the security of their codes was also shared by the German Army cryptographers. One of them was lieutenant Fritz Nebel who served in the Great General Staff at Spa. Nebel achieved the Abitur (German university entrance qualification), specializing in Greek and Latin, but when he joined a telegraph battalion, he had no prior cryptography training. Nevertheless, he was asked in early 1918 to create a new and highly secure field cipher, which Germany needed for its coming Spring Offensive. Nebel designed the ADFGX cipher, which he later extended by adding one more symbol (ADFGVX).

In the late 1960s Fritz Nebel met for the first time with Georges Painvin, his former French opponent, at a historical meeting of former cryptologists in Paris. On that occasion, Nebel was shocked when the French cryptanalyst described to him how he had cracked the ADFGVX cipher, including an intercepted German message in May of 1918 containing critical strategic information about German plans, the famous "Radiogramme de la Victoire". Painvin's success

¹The German term "Nachrichten" is ambiguous. "Nachrichten" is used either in the context of intelligence, news, and information, or in the context of signals and communications. The intelligence officers of Abteilung IIIb, attached to the Army Corps, called themselves "Nachrichten-Offiziere", but often so did signal officers, who dealt mainly with military communication systems.

had enabled the Allies to stop the German offensive and to take back the initiative on the Western Front. Nebel later confessed that he was "neither a mathematician nor an engineer" when he developed the codes [DL98, p. 20]. His confession confirms Kahn's harsh verdict on German cryptology in WWI.

6.2 James Rives Childs and Allied Cryptanalysis

The French successes with German codes in 1914 were not a coincidence. Since their defeat in the 1870-71 Franco-Prussian War, French military intelligence had worked hard on cracking German codes. This knowledge enabled the French Deuxième Bureau to intercept and decipher German encrypted messages from the beginning of the war. The French high command was able to obtain the enemy's order of battle and to learn of his plans to attack Paris, and they were able to stop the German offensive in the West, at the famous "miracle of the Marne".

After this success the French cryptanalysts joined forces with their British allies and from 1917 with their American counterparts. They openly exchanged intercepts, decrypts, knowledge, and methods on a continuous basis. On the US side the Radio Intelligence Division, under the command of Lieutenant Colonel Frank Moorman, was attached to the General Staff of the American Expeditionary Forces (AEF). It included the Radio Interception Section and the Code Solving Section, headed by William F. Friedman.

One of Friedman's officers was James Rives Childs. Childs was born in February 6, 1893 in Lynchburg, Virginia. During 1909-1911 he studied at the Virginia Military Institute and from 1912, at Randolph-Macon College where he obtained his bachelor's degree. After receiving his master's degree from Harvard in 1915, he joined the US Army, and served in the American Expeditionary Force. He was assigned to G.2 A.6, the radio interception service, as the liaison officer with the French and British codebreaking services. During this period, he worked closely with George Painvin, learned from his methods which he successfully applied to Eastern Front ADFGVX traffic. After the war he developed a general solution for the ADFGVX cipher. Childs received the Medal of Freedom for his work cracking German codes [Lof14].

After the war, Childs entered the U.S. Foreign Service. During World War II, he was the chargé d'affaires at the US Legation in Tangier, Spanish Morocco, where he helped saving 1200 Hungarian Jews from the Holocaust by securing Spanish visas for them. During his 30-year career as a US diplomat he held several ambassadorship posts in the Middle East. Childs passed away on July 15, 1987, in Richmond, Virginia. He wrote numerous books and articles, about his service as an ambassador, about Casanova, and about his work on deciphering German ciphers. His papers are held in a special collection at the University of Virginia.

In 1919, while still in Paris, he wrote "The History and Principles of German Military Ciphers, 1914-1918" [Chi19]. This manuscript was not published, and copies are available at the Randolph-Macon College - University of Virginia Library as well as in the Kahn Collection at the National Cryptologic Museum in Fort Meade, Maryland. Interestingly, a German version of the manuscript, translated in 1969 at Bad Godesberg, is also available at the Library of the German Armed Forces [Chi69]. The manuscript contains an introduction by David Kahn. In 1935 the US War Department published a report named "German Military Ciphers from February to November 1918". This report was declassified by NSA in 2014 [Chi35]. Though the document contains reports from several authors, it underlines Childs' authorship.

In 1934, the US War Plans and Training Division published the "General Solution of the ADFGVX Cipher System", also recently declassified and available on the NSA site [Fri34]. In 2000, the report was published by Aegean Park Press [Chi00]. This later edition includes the 460 enciphered radiograms which we have solved and analyzed in this article. The editor states that those WWI intercepted German messages were furnished by Childs "some 20 years ago" (i.e. around 1980) [Chi00].

6.3 Reading German Communications from Romania

In this section we present some of the key intercepts and decrypts of ADFGVX traffic related to Romania. We start with an overview of the political and military situation towards the end of the war, as well as Germany's communications infrastructure in that country. We also present the callsigns used by the relevant radio stations. While some of the callsigns were identified by Childs, an analysis of the traffic (similar to traditional Traffic Analysis) as well as external sources were required to recover the remaining callsigns.

On May 7th, 1918, following its defeat by Germany, Romania signed the "Peace of Bucharest" treaty. Under the clauses of this treaty, most of the Romanian army had to be demobilized, and the country was to be occupied by German forces. Field Marshall August von Mackensen was assigned as the head of the German occupation army. Its HQ, the Oberkommando Mackensen (OKM), was initially in Bucharest, with callsign UKS. The 11th Army (AOK 11, callsign ZÖN²) was also assigned to this area. Mackensen later transferred his personal headquarter to the palace of Pelesch, 120 kilometers north of Bucharest. Mackensen seemed to appreciate modern communications technology - telephone and telegraph. "The war has perfected all these facilities. We can talk with headquarters as two people sitting in front of each other", he remembered [Sch96, p. 149].

A vast railroad network connected occupied Romania to the German Empire, and to the port of Constanza (callsign COS) on the Black Sea. Another port of strategic and logistical importance for Germany was the Ukrainian port of Nikolajev (callsign NKJ). Every day 100 wagons with petroleum left for Germany which badly needed it for its war machinery.

The outcome of the war, however, depended mainly on developments on the Western Front. The 1918 German spring and summer offensives eventually failed, partly due to Allies cryptanalytic successes. On September 29, 1918 the Kaiser's Supreme Army Command asked its government to seek an armistice with the Entente, to avoid a complete military defeat and the occupation of Germany. In October, the Austria-Hungary empire started to disintegrate into smaller nations. Some of them, like Bulgaria, left the alliance with the Central Powers (Germany and Austria-Hungary), and signed an armistice with the Entente. On October 31, 1918 the Hungarian Government terminated the union with Austria, and on November 3, Vienna signed an Armistice with the Entente. That same day, social and political protest in Germany against the war culminated into a mutiny of the German Navy at Kiel, and spread all over the German Empire and the occupied countries.

On November 3, 1918, a 13-part message was intercepted by the British, and decrypted by Childs. Two of the parts are missing and some of the letters are garbled. It contains "the entire plans by which General Mackensen proposed to retreat from Romania" [Chi35, p. 14].

²In Childs' transcriptions there are also the versions ZO'N and ZON.

Part	Reconstructed German text	Translation	Cryptogram starting with
1	BEURTEILUNG DER LAGE BISHER MUSSTE DAMIT GERECHNET WERDEN DASS FEIND MIT DEN BEI VIDIN LOEPALANKA UND IN GEGEND RUTSCHUK IN VERSAMMLUNG BEGRIFFENEN KRAEFTEN EINEN DONAUUEBERGANG VERSUCHEN WIRD MIT DEM ZIEL DIE BAHN ORSOVA CRAIOVA ZU UNTERBRECHEN UND AUF BUKAREST VORZUSTOSSEN	Evaluation of the situation Until now it had to be expected that the enemy would try to cross the Danube with the troops assembling at Vidin Lompalanka and near Drutschuk with the goal to cut off the railways between Orsova and Craiova and to push forward to Bucharest	GXAAG XVGFF
2	[Original cryptogram missing. Translation from Childs[Chi78, p. 212]]	Since November 1, 1918, it appears that the Serbian armies, together with three French divisions, are engaged in an advance toward Belgrade-Semendria, and the intended attack at Vidi and Lompalanka seems to have been abandoned.	
3	MIT DEM AUFMARSCH STAERKERER KRAEFTE AN DER DONAU SUEDLICH SVISTOV RUSTSCHUK MUSS BESONDERS NACH FRIEDENSSCHLUSS DER TUERKEI WEITER GERECHNET WERDEN [Chi00, p. 113 A]	It has to be expected the deployment of stronger forces at the Danube south of Svistov Rustschuk specially after the peace agreement of Turkey.	DDAGA AXGGG
4	ES IST SOMIT DURCHAUS WAHRSCHEINLICH DASS DIE DURCH FRANZOSEN VERSTAERKTEN SERBISCHEN ARMEEN DEN UEBERGANG UEBER DIE [Chi00, p. 114 A]	Hence it is most likely that the Serbian armies, reinforced by the French, intend to cross the	VXFXD GVDGX
5	DONAU BEI BELGRAD SEMENDIA UND DEN EINMARSCH IN SUE SUE UNGARN BEABSICHTIGEN WAEHREND DIE AUFGABE DER SUEDL	Danube near Belgrade Semendia and the into southern [repeat: southern] Hungary. While the task of the	VGGGV XDVGD

Part	Reconstructed German text	Translation	Cryptogram starting with
6	SVISTOV RUSTSCHUK AUF-MARSCHIERENDEN FRANZOES ARMEE OFFENSIVE RICHTUNG BUKAREST BESTEHEN BLEIBT IM ZUSAMMENHANG MIT DIESER	French Army marching south of Svistov Rutschuk remains the offensive toward Bucharest. In conjunction with	GADAG GXDGV
7	OPERATION IST ES NICHT AUSGESCHLOSSEN DASS RUMAEN KRAEFTE AUS DER MOLDAU DURCH DEN TOELGYES GIMES UND OITOS PASS IN	this operation it cannot be ruled out that the Romanian forces coming from Moldavia over the passes of Toelgyes, Gimes and Oitos [will invade Transylvania]	XAVGG AGFFA
8	SIEBENBUERGEN EINRUECKEN DADURCH WERDEN DIE RUECKWAERTIGEN VERBINDUNGEN DES BESATZUNGSHEERES DIE BISHER NUR IN FOLGE	Thus the lines of communications in the rear of the Occupation Army, which have up to now as a result of	GAXVA FAVFG
9	(9th part missing) [Chi00, p. 114-115]		
10	ANGRIFF BEDROHT UND DIE WEITERE BESETZUNG DER WALACHEI WIE IN O K 2 RM1 A NR 11161 AUSGEFUEHRT ZWEIFELLOS UND IM HINBLICK	is threatened with attack and the further occupation of the Walachia as laid down in order from OK (Headquarters) 2 RM NR 11161. Without doubt and with regard to	VAGFD DFXGA
11	AUF MUNITIONS VERPFLEGUNGS UND KOHLEN VORRAETE UN-DURCHFUEHRBAR FALLS DER ALLGEMEINE WAFFENSTILLSTAND NICHT IN ABSEHBARER	the ammunition, food, and the coal stocks, it is unfeasible. If the general Armistice does not become effective in the foreseeable future	GDVGA VFGG
12	ZEIT EINTRITT WIRD DESHALB VORGESCHLAGEN DAS BESATZUNGSHEER SOFORT AUS RUMAENIEN HERAUS ZU ZIEHEN UND GEMEINSAM MIT DEN DEUTSCHEN	it is suggested that the Occupation Army be withdrawn at once from Romania and together with the German	A-VDA GGXGG
13	[Many symbols missing. Some of the translation from Childs [Chi00, p. 213]]	units of the first Army to start the march to Upper Silesia through Hungary. Approval is requested. (Signed) K.M. I A GR-OP	GVVFG -GFGD [Chi00, p. 116]]

Table 3: Message from Mackensen HQ to the Supreme Army Command – November 3, 1918

This long message was sent from Mackensen HQ (OKM, using callsign UKS) to the Supreme Army Command (OHL, callsign LP) via Nikolajev (callsign NKJ). Its decryption provided the Allies with an insight into Mackensen's evaluation of the situation and his planning. According to Kahn, the decrypted message was immediately handed over to the French Supreme War Council [Kah96, p. 339].

On November 9, the Kaiser was forced to abdicate, and Germany became a republic. Two days later, the German government signed the Armistice at Compiègne and started to withdraw its armies from the Western Front. The war was over but hundreds of thousands of German soldiers were still deployed in the Eastern Front, cut off from their homeland. Due to a shortage

of coal, wagons, and engines, an organized evacuation could not be planned. The situation became critical as the Entente imposed a timetable which could not be met.

On November 25, 1918, at 16:30h, OKM sent to OHL in Berlin another message, also intercepted and decrypted by Childs. Mackensen informs the Supreme Army Command, the War Ministry, and the new Government at Berlin that the French commanding officer in the region, general Berthelot, has notified him that general Foch refuses to apply the conditions of the Armistice to the Southeastern Front, and demands the immediate disarmament and internment of Mackensen's troops. Mackensen is awaiting a decision from OHL on the matter before confirming the reception of Berthelot's message. He also proposes that disarmament be performed by Hungary rather than by Entente countries, and that internment take place in Hungary, to guarantee appropriate food supply for the troops. He is concerned that the situation could deteriorate into a "mass exodus with indescribable misery and open revolt" [Chi00, p. 221 A].

While waiting for a final decision, Mackensen is getting more and more concerned about communications. He cannot use Nikolajev (NKJ) anymore as a relay station, and he communicates with Berlin via the Breslau station (BSL). On November 27, 1918 OKM requests BSL "to staff the station with very good operators" because radio is now "the unique secure communication with the homeland" [Chi00, p. 224 A].

On November 28, the Allies intercept a message from Berlin (LP) via Breslau (BSL) to OKM (ZON) with instructions "to avoid every kind of struggle and to accept the French demands. Further orders will follow" [Chi00, p. 227 D]. Final instructions arrive November 29, 1918 and this message is also intercepted and decrypted by Childs. The fate of Mackensen's army is sealed (see Table 4).

Part	Reconstructed German text	Translation	Cryptogram starting with
1	SOLDATENRAT DER ARMEE MACK- ENSEN KAMERADEN TROTZ ALLER BEMUEHUNGEN DER IN BETRA- CHT KOMMENDEN SITUATIONEN IST ES UNS NICHT GELUNGEN DIE ENTENTE VON IHREM UN- GERECHTFERTIGTEN STAND- PUNKTE ABZUBRINGEN	To comrades of the Soldiers Coun- cil of the Mackensen Army. De- spite all efforts regarding the com- ing situation we have failed to con- vince the Entente to renounce its unjustified point of view.	FVVVA GDVXF
2	SIE BEHARRT AUF EURER IN- TERNIERUNG UND BETRACHTET DIE AUFRECHTERHALTUNG DES ABMARSCHBEFEHLS ALS EINEN BRUCH DES WAFFENSTILL- STANDES DER IHR DAS RECHT GIBT IN DEUTSCHLAND EINZU- MARSCHIEREN	It insists on your internment and views the order to march as a vio- lation of the Armistice which gives it the right to invade Germany.	VVFFD FVFXG
3	ES IST UNSER BEMUEHEN EURE INTERNIERUNG SO ZU GESTAL- TEN DASS IHR WEDER NOT NOCH MANGEL LEIDET UND NACH DEM MOEGLICHST RASCHEN FRIEDENSSCHLUSS GLUECKLICH UND GESUND IN DIE HEIMAT ZU- RUECKKEHREN COENNT [Chi00, p. 226 C]	Our efforts are to arrange your in- ternment so that you do not have to suffer deprivation and that you can safely come home as fast as possible	XDFFV FVDGF
4	HALTET RUHE UND ORDNUNG DAMIT NICHT DURCH INNERE AU- FLOESUNG IHR ALLE VERLOREN GEHT WIR HOFFEN EUCH ALLE RECHT BALD GESUND IN DEM NEUEN DEUTSCHLAND WIEDER ZU SEHEN VOLLZUGSRAT DER A UND S R ABTE GROSSBERLIN SOLKENBUHR MUELLER [Chi00, p. 227 A]	Maintain law and order so that you will all not get lost by inter- nal dissolution. We hope to see you all again, soon and healthy, in the new Germany. Execu- tive Council of the W[orkers] and S[oldiers] Councils, Dept. Great Berlin Solkenbuhr, Müller.	FFFFV XGVXF

Table 4: Final instructions to the 11th Army in Romania – December 2, 1918

The last message about Romania in Childs' collection dates from November 30, 1918. An unidentified officer named Papp, using callsign CV, asks the 11th Army to "send the laws of internment in time" [Chi00, p. 230 E]. In December the Field Marshal enters Hungary with his remaining troops. He is interned in a palace with comfortable conditions and guarded by French troops, while most of his former soldiers are sent back to Germany.

6.4 The German November Revolution

On October 28 a military and civil uprising against the war and the Imperial Government starts at Kiel, the main port of the Imperial Navy. The crews of the Imperial Navy refuse to obey order for a last and futile attack on the British Navy. At first, the military authorities attempt to op-

press the Kiel mutiny, without success. On November 3 the mutiny turns into an overt rebellion which begins to spread all over the German Empire and the occupied countries. The German government and the Supreme Command of the combined armies (Oberste Heeresleitung, OHL) are still trying to contain the spread of the rebellion. As part of this effort, OHL needs to secure the communications infrastructure to avoid the takeover of radio stations by the rebels.

At 13.31 on November 6 the OHL Chief Signal Officer (Nachrichtenchef) sends an order marked with "high urgency" from the main military radio station at Berlin (callsign LP) to the army radio stations at Constantinople (YZ) and Nikolayev (NKJ), and via them to all radio stations in the Black Sea Area, including Romania, Georgia, and Turkey [Chi00, p. 140 A]. All radio stations should be taken over immediately by an officer who should "keep a firm grip on them". The officer should prevent any radio messages from Kiel from being recorded and distributed. Any such previously received texts should be confiscated and destroyed. The radio station at Nikolayev forwards this message to the station ASO (probably Tiflis in Georgia), and the first part of this transmission is intercepted and decrypted by Childs on November 7th [Chi00, p. 146 A].

On November 8 the signal officer informs the General Staff at Spa in Belgium (callsign SY) that the German military radio station in Warsaw has "surrendered to the Soldiers' Council" [Chi00, p. 151 C]. One day later, signal officer Seifert (his name also appears in other messages) informs SY that from 22:00h the radio station will be under the control of the Soldiers' Council, but that he still has a "firm grip" on the station and its personnel [Chi00, p. 168 A].

On November 9, the German Republic is proclaimed in Berlin. On the same day, a new message is intercepted and decrypted by Childs. It describes how the Revolution is spreading from the Baltic Sea to the western parts of the German Empire. The signals division (N4) informs the HQ at Spa that the Soldiers' Council of Cologne is trying to take over the local radio station [Chi00, p. 168 B]. Cologne and its infrastructure were of strategic importance for both the Navy and the Army. Still trying to maintain control over communications, the Supreme Command sends an order - also intercepted by the Allies - to stop using the cable network (phone and telegraph), and instead use only radio transmission and army ciphers. All messages should be confirmed twice [Chi00, p. 168 C and D].

Undoubtedly, the November Revolution had a major impact on Germany's willingness and haste to agree to an armistice. The Allies were able to follow the travel preparations of the German delegation to the negotiations, and to identify the unit in charge of this operation. A message from November 10, 22:57h, for instance, refers to captain von Weiher, commander of a company of the 1st Bataillon of the 1st Foot Guards Regiment. Since the Napoleonic Wars all princes of Prussia served in this Regiment, whose commander traditionally was the King of Prussia. The intercepted message indicates that the "Erste Garde Regiment" is responsible for organizing the transport "by train or car of the members of parliament to Spa". They are scheduled to arrive to Berlin on November 11 at 15:00h, after having signed the Armistice [Chi00, p. 172 C and p. 175 A].

On November 11, at 19:06h, the military radio station in Berlin (LP) sends the details of the armistice to Nikolayev (NKJ) with instructions to forward them "by all means" to the Mediterranean Division (MMD) and to the Special Command of the Navy Operational Command (SKL). This message was probably sent quite late as it states that the armistice should take effect at noon on the same day. "Every act of war has to be stopped" and the naval forces should be confined to their bases. Submarines also have to be informed - using a ciphered message - that the armistice had come into effect. Weapons should be used only for self-defence. Finally LP informs that permission has been requested from neutral governments for safe passage of German ships over neutral territorial waters [Chi00, p. 178 A and p. 178 B].

One of the conditions of the Armistice was that the complete demilitarization from German troops of the region between the Rhine river and Germany's western border. As a result, the Army GHQ moved to Kassel. However, the revolution also reached this city. On November 10, the lower army ranks disarm their officers. A Workers and Soldiers' Council is established, and a red flag hoisted over the city's arch of triumph. An officer is shot while trying to tear down the flag. Those events are also reflected in the intercepted ADFGVX traffic. In a message from November 12 at 22:09h, a certain lieutenant Bauman[n], probably the signal officer in charge of

a radio station, is ordered "to be ready to move to Kassel via a M[ilitary?] transport, as soon as possible" [Chi00, p. 180 E].

On November 13 the Soldiers' Council in Warsaw asks the radio station at Nikolayev whether the rumors that field marshal Mackensen is marching into Warsaw with his troops are true. Mackensen was known as a strong supporter of the monarchy, and the Soldiers' Council feared that Mackensen would cross Poland with his 170, 000 soldiers on their way from Romania to Germany [Chi00, p. 181 A].

6.5 A Code within a Code

Three messages, divided into four cryptograms, contain enigmatic references to unidentified people or entities, such as 'Dictator' and 'Herkules'. The first message was sent on November 4, at 14:05h, from Berlin (LP) via Nikolayev to Bucharest. The Kriegsamts (the War Office) informs Mackensen's economic staff firstly that "the gold transport has arrived", and that "Herkules" rejects again any responsibility for new emergency calls from "Andreas" [Chi00, p. 122 C].

On the same day, at 19:56h, a second message was sent from the Navy radio station at Eilvese (callsign OUI) to Osmanié (OSM), the major radio station at Constantinople. It consists of three parts, which appear on separate pages in Childs's book. The first part on page 120, while the second and third parts are on page 123 [Chi00, p. 120 B and p. 123 A]. The sender is the supply department of the German Empire (Reichsversorgung). It informs the "Feigenhaus" ('house of figs') in Constantinople that "Diktator" has given the instruction "not to make emergency calls to [sic] Andreas", and that the Reichsversorgung has again contacted "Herkules". This enigmatic person or institution rejects any further responsibility "for new calls for help". "We cannot make new emergency calls" states the Reichsversorgung and orders "Feigenhaus" to act accordingly. In addition Feigenhaus should report via telegraph "what it is planning to do regarding the political situation, which keeps changing."

The third message is from November 15, at 16:50h, and is signed by the supply department and by the Admiralty Staff. It was sent by the main radio station at Berlin (LP) to the radio station with a the new callsign XYZ, probably a German Army radio station in Constantinople. According to this message, "Herkules" has informed Berlin, that on November 12 several steamers from "Operette" have reached Nikolayev with nearly 10 tons of liver and cacao on board. "The cargo remains under military guard" because of transport problems. The economic staff is required to gather more information on that issue, and to inform the supply department about OKM's intentions for this shipment [Chi00, p. 199 A].

We may only speculate about the identities of the persons and organization. The "Diktator" could simply refer to von Hindenburg, the head of the OHL. "Andreas" is possibly a codename for Georgia, a newly created state, where Germany had a military presence and strong economic interests often conflicting with the interest of its ally, the Ottoman Empire. "Herkules" could be a codename for either OHL or Naval Command. "Feigenhaus" could be the supply or the economic staff of the German Army in Constantinople, and "Operette" the name of a port in the region. Still, it is not clear why codenames were used in those messages, while most of the other messages did not use them.

6.6 Hagelin mentioned in an ADFGVX Message

Among the decrypted messages we have found a message mentioning Hagelin, most probably Karl Wilhelm Hagelin, the father of the famous cryptographer Boris Hagelin. The latter was born in Adshikent, Azerbaijan in 1892, where his father was director of the „Naphtaproduktionsgesellschaft Gebrüder Nobel". The company had several oil production facilities in Baku and its headquarters in Saint Petersburg. Karl Wilhelm was a member of the management staff of Emanuel Nobel, the "Swedish Rockefeller" and the nephew of the famous Alfred Nobel, the inventor of dynamite. After Alfred's death, Emanuel implemented his will and established the Nobel Prize foundation. After the Bolshevik revolution, the Hagelin family had to leave Russia. With the help of his father and Emanuel Nobel, Boris Hagelin joined Arvid Damm's Aktiebolaget

Cryptograph and later turned it into one of the most successful vendors of encryption machines.

On October 5 1918 , at 03:23h, the Allies intercepted a message mentioning Hagelin [Chi00, p. 64 A]. The sender is probably the quartermaster of the German Army in the East, the Ostarmee. The message was sent via the Nikolayev radio station (NKJ). The receiving station is unknown since the beginning of the message is missing. The message is for captain Benedikt, the representative of the Austria-Hungarian Army Command in "Atom". The officer should deliver the telegram to "Gebrueder Nobel Atom" and make sure that the response is sent back to the Ostarmee. Details are requested about stocks of "naphtha production of all sorts in our possession and with other Atom companies, how much has been delivered and the same information about Novorossiysk, if known - Hagelin". Novorossiysk is a Black Sea port in the Caucasus region, from which oil supplies from Baku were shipped. The name 'Hagelin' appears near the signature, but the context is not clear, whether the information should be sought from Hagelin, or whether the sender is not sure whether or not captain Benedikt knows Hagelin.

— v NKJ (3.23 a.m., October 5th)
(Beginning lost)

Part	Reconstructed German text	Translation	Cryptogram starting with
1	HPTM BENEDIKT VERTRETER KUK AOK IN ATOM ATOM ERSUCHE DRINGENDE ZUSTELLUNG FOLGENDEN TELEGR U VERANLASSUNG DASS ANTW TELEGR ANHER UEBERMITTELT WIRD GEBRUEDER NOBEL ATOM DRAHTET VORRAETE	Capt. Benedikt, representative of the Austria-Hungarian Army Command in Atom Atom requesting urgent delivery of the following telegram and that answer be transmitted telegraphically to here Nobel Brothers Atom. Telegraph stocks of	GDDD FGFAF
2	NAPHTA PRODUKTION SAEMTLICHER SORTEN BEI UNS UND ANDEREN FIRMEN ATOM FERNER WIEVIEL BIS JETZT GELIEFERT DIESELBEN DATEN BETREFFEND NQVOROSSIYSK FALLS SOLCHE IHNEN BEKANNT HAGELIN OSTARMEE O QU 12059	naphtha production of all sorts in our possession and with other companies Atom, how much has been delivered and the same information about Novorossiysk, if known - Hagelin Ostarmee Quartermaster	AVDFF VDVVG

Table 5: Message mentioning Hagelin - October 5, 1918

7 Conclusion

The success by France, the UK, and the US in breaking almost all German ciphers, and the ADFGVX in particular, exemplifies the failure by Germans to implement and deploy secure communication systems. Surprisingly, it seems that the German military and politicians did not draw the full lessons from their failures on the cryptological front. Most nationalist Germans, and later Nazi Germany, considered the defeat to be the result of internal "betrayal" and of the November Revolution. Under the Nazi regime, there were a multitude of cryptanalytic agencies, but there was no centralized entity directing and coordinating the cryptological efforts of the different units, and the mistakes of WWI were often repeated in WWII. In contrast, in WWII, cryptanalytic efforts by the UK and the US were highly centralized, with codebreaking experts such as Friedman playing a central role in the development of cipher systems for the US military.

The interception of ADFGVX messages, most of which were successfully decrypted, demonstrate the advanced capabilities of the Allies to monitor German communications and to break into their most advanced ciphers. The Allies gained in-depth insight about political and military

developments inside Germany and on both war fronts. Further study is needed to assess to what extent the intelligence gathered from ADFGVX Eastern Front was used by the Allied powers for shaping their policy towards Berlin, and for military planning in that front.

On the technical side, ADFGVX was the most challenging cipher of its time. Painvin's successes, based on special cases (similar endings or beginnings), were quite impressive, taking into account that they were achieved under huge time pressure and during Germany's last major offensives on the Western Front. Childs successfully mastered the techniques he learned from Painvin, and was able to decipher a large part of the Eastern Front ADFGVX traffic. Although after the war Childs also developed a more general solution for the cryptanalysis of ADFGVX, some of the messages still could not be deciphered. Fortunately, the intercepted messages were preserved by Childs. Using our new ciphertext-only method which is based on a divide-and-conquer approach and an hillclimbing algorithm, we were able to decrypt almost all the messages. Those decryptions, together with historical research combined with classical "traffic analysis", provide a unique strategic political and military insight into the last stages of the war in this "forgotten front".

About the Authors

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Ingo Niebel is a historian and freelance journalist living and working in Germany. He received his PhD from the University of Cologne in 2012. Since 1996, he is also a member of Eusko Ikaskuntza, the Society of Basque Studies. He has published several books in German and Spanish about the Basque Country and intelligence during the Spanish Civil War (1936–1939). During his research he uncovered original Civil War telegrams encrypted using the Spanish Strip Cipher, that could not be read as the encryption keys had been lost. He was part of a team led by Professor Christof Paar, who has the Chair for Embedded Security at Ruhr University Bochum, Germany. The team successfully decrypted those historical messages [SLP⁺16].

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Professor Arno Wacker is the head of the research group "Applied Information Security" (AIS) with the University of Kassel in Germany. He is also the lead of the open source project CrypTool 2.0 [KKWE14, Cry16] and member of the steering group of MysteryTwister C3. His main research interests are modern security protocols for decentralized distributed systems, e.g. for volunteer computing scenarios, and computerized cryptanalysis of classical ciphers. Additionally, he teaches classical and modern cryptology in classes at the University of Kassel and at special workshops for pupils at local schools.

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